

# Barret Schloerke

Senior Software Engineer, Posit

`code[c("architect", "manager", "philosopher")]`

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## Summary

Software engineer at Posit, where I build and maintain tools in the Shiny ecosystem for both R and Python, including shiny, py-shiny, shinytest2, shinylive, plumber, and reactlog. Lately that means OpenTelemetry for Shiny, LLM-powered apps, and custom UI with React. PhD in Statistics from Purdue University, focused on large data visualization.

## Experience

**Senior Software Engineer**, [Posit](#) · Remote Jan 2022 – Present

- Core developer of Shiny for Python, a framework for building interactive web apps straight from Python.
- Architected Playwright testing controllers for every Shiny for Python UI component, run by 1000+ pytest tests across parallel CI workers.
- Built shinylive for R and Python, which export Shiny apps to static sites that run in the browser with WebAssembly.
- Lately: OpenTelemetry for Shiny, LLM-powered apps (querychat, shinychat), and custom Shiny UI with React (shinyreact).

**Software Engineer**, [RStudio \(now Posit\)](#) · Remote Jan 2018 – Dec 2021

- Core developer of Shiny, an R package for building interactive web apps straight from R.
- Architected and automated nightly testing of 150+ Shiny apps with GitHub Actions on Mac, Windows, and Linux across 5 R versions (shinycoreci).
- Converted Shiny's core JavaScript to TypeScript.
- Maintained plumber (R web APIs), reactlog (reactivity visualizer), shinytest2, and dozens of packages in the Shiny ecosystem.

**Tool Development Team**, [Gates Foundation: HBGDki](#) · Remote Apr 2016 – Dec 2018

- Produced custom summary statistics for 75+ datasets to feed visualization apps.
- Kept data secure while coordinating with five other data scientists.
- Created and maintained template R packages for standardized package development.

**Researcher**, [DARPA XDATA and D3M grants](#), [Purdue University](#) · West Lafayette, IN Sep 2016 – Dec 2017

- Worked remotely with Hafen Consulting, Kitware, and KnowledgeVis on a web interface for subject matter experts to explore machine learning pipelines.
- Lead point of contact for integrating services with other working groups.

**Software Engineer**, [Metamarkets](#) · San Francisco, CA Jan 2011 – Jul 2012

- Built interactive D3 visualizations on DVL, a reactive data visualization library, in CoffeeScript.
- Wrote Node.js front-end servers: automated deploys, server clusters, and company / user management.
- Cut site startup time and trimmed data queries to reduce server load.

**Data Scientist Intern**, [Novartis Pharmaceutical R&D](#) · Basel, Switzerland May 2008 – Aug 2008

- Built a Gantt chart tool showing employee availability for management.
- Wrote a web tutorial of common graphics as reusable plotting layers.

## Education

**PhD, Statistics**, [Purdue University](#) · West Lafayette, IN 2013 – 2017

- Advisors: William Cleveland and Ryan Hafen. Emphasis: large data visualization.
- National Science Foundation Graduate Research Fellow (2012 – 2017).

**MS, Mathematical Statistics**, [Purdue University](#) · West Lafayette, IN 2013 – 2014

**PhD program, Statistics**, [Rice University](#) · Houston, TX 2012

- Advisor: Hadley Wickham.

**BS, Computer Engineering**, [Iowa State University](#) · Ames, IA 2006 – 2010

- Statistics graphics research with Dianne Cook and Heike Hofmann.

## Open source software

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2,300+ merged pull requests across 160+ GitHub repositories since 2018. Packages I maintain or author, by merged PRs:

**shiny** Python · author · 350+ PRs · A web development framework for Python.

- Agent Skills (2026): Skills shipped in the package so coding agents know how to write Shiny apps.
- Test mode (2026): `test_server()` and exported test values for fast tests without a browser.
- Session lifecycle (2026): `session.destroy()`, a soft teardown when a session closes, and `session.allow_reconnect()`.
- Release train skill (2026): An Agent Skill that walks a py-shiny release from prep through PyPI, conda-forge, and shinylive.
- OpenTelemetry (2026): Traces and logs for sessions, reactivity, and outputs.
- Bookmarking (2025): Save and restore an app's state in the URL or on the server.
- Data frames (2024): `render.data_frame` with editing, selection, sorting, filtering, and styles.
- Output renderers (2024): `shiny.render.renderer`, a base class for writing your own output renderers.
- API reference (2023 – 2026): The quartodoc function reference and its runnable examples.
- UI components from bslib (2023 – 2024): Ported sidebars, cards, accordions, value boxes, tooltips, and popovers from R's bslib.
- Playwright testing (2023 – 2024): `shiny.playwright` controllers for every UI component, and the `shiny.pytest` plugin.

**shinyreact** R, Python, TypeScript · maintainer · 170+ PRs · Client-Side 'React' Interface for 'shiny'

- TypeScript: npm package (2026): A dual build published as `@posit-dev/shinyreact`, usable with no server-side JS.
- TypeScript: Wire protocol (2026): One versioned protocol, checked by a handshake, shared with the R and Python packages.
- Python: Agent Skills (2026): Skills for building and converting apps with coding agents, shipped in the package.
- R: Agent Skills (2026): Skills for building and converting apps with coding agents, shipped in the package.
- R: Initial development (2026): Ported the Python API to R, then CRAN checks, a pkgdown site, and webR builds for 0.1.0.
- Python: Renderer dependencies (2026): Find and serve the HTML dependencies of Shiny outputs used inside React.
- Python: `ui.tsx` apps (2026): Write the whole page in React and serve it from Python.
- R: `ui.tsx` apps (2026): Write the whole page in React and serve it from R.
- TypeScript: React hooks (2026): `useShinyInput` / `useShinyOutput` hooks, and `ShinyOutput` for classic Shiny outputs.
- Python: Testing (2026): Playwright e2e tests, WireTap for wire payloads, and in-memory `test_server()`.
- Python: Initial development (2026): Began as `shinyjson`, a JSON spec renderer, then turned React-first and dropped `json-render`.

**GGally** R · maintainer · 150+ PRs · Extension to 'ggplot2'

- S7 (2025): Moved `ggmatrix` to S7 for `ggplot2` 4.0.
- Discrete stats (2020): Count, cross, table, and statistic plots for GGally 2.0.
- `ggduo`, `ggnostic`, and friends (2016 – 2017): Plot matrices for two groups of variables, model diagnostics, and time series.
- `ggpairs` and `ggmatrix` (2014 – 2017): The generalized pairs plot and the plot matrix behind it.

**shinytest2** R · maintainer · 140+ PRs · Testing for Shiny Applications

- Migration from `shinytest` (2022): `migrate_from_shinytest()` converts existing `shinytest` tests.
- Screenshots (2021 – 2023): Consistent screenshots, compared with a kernel density smoother to ignore stray pixels.
- Test recorder (2021 – 2022): `record_test()` writes a test while you click around the app.
- `AppDriver` (2021 – 2022): Drive a Shiny app in headless Chrome: set inputs, click, and snapshot values.

**shiny** R, TypeScript · author · 140+ PRs · Web Application Framework for R

- R: MCP Apps (2026): Serve Shiny apps as MCP Apps, with author tools for LLM clients.
- R: Session lifecycle (2026): `session$destroy()`, a soft teardown when a session closes, and clean state on reconnect.
- R: OpenTelemetry (2025): Traces and logs for sessions, reactivity, and outputs through `{otel}`.
- R: Quosures (2021): `exprToFunction()` and `installExprFunction()` accept quosures, including inject()-ed ones.
- TypeScript: TypeScript (2021 – 2022): Ported `shiny.js` to TypeScript, moved the build to esbuild, and published the types.
- R: `devmode()` (2020 – 2021): One switch that turns on the options you want while developing an app.
- R: `testServer()` (2020): Test server functions and modules in memory against a mock session.
- R: Plot brushing (2018): Plot click and brush events carry CSS and image coordinates, and brushing works in WebKit.
- R: reactlog integration (2018 – 2020): Record every reactive event so `reactlog` can draw the reactive graph.

[plumber](#) R · maintainer · 130+ PRs · An API Generator for R

- `pr_*`() API (2020): The pipe-friendly router API and consistent R6 methods of plumber 1.0.
- Docker (2020 – 2021): Docker images for running plumber APIs.
- Async (2019 – 2020): Promise and future support for long-running endpoints.
- Serializers and parsers (2019 – 2020): Body parsers chosen by content type, and serializers for many output formats.
- OpenAPI docs (2018 – 2020): OpenAPI v3 specs and pluggable docs pages (Swagger, RapiDoc, Redoc).

[learnr](#) R · author · 70+ PRs · Interactive Tutorials for R

- Exercises (2019 – 2020): Safer exercise evaluation, checker functions, and clearer errors.
- Quiz questions (2019 – 2020): Questions rebuilt as Shiny modules, with a custom question API.

[promises](#) R · maintainer · 40+ PRs · Abstractions for Promise-Based Asynchronous Programming

[leaflet](#) R · author · 40+ PRs · Create Interactive Web Maps with the JavaScript 'Leaflet' Library

[reactlog](#) R · maintainer · 40+ PRs · Reactivity Visualizer for 'shiny'

[shinyloadtest](#) R · maintainer · 30+ PRs · Load Test Shiny Applications

[htmltools](#) R · author · 30+ PRs · Tools for HTML

[shinylive](#) R · maintainer · 20+ PRs · Run 'shiny' Applications in the Browser

[shinyswatch](#) Python · author · 20+ PRs · Bootswatch + Bootstrap 5 themes for Shiny.

[chromote](#) R · author · 17 PRs · Headless Chrome Web Browser Interface

[sortable](#) R · author · 12 PRs · Drag-and-Drop in 'shiny' Apps with 'SortableJS'

[trelliscopejs](#) R · author · 11 PRs · Create Interactive Trelliscope Displays

[lmmr](#) R · author · 9 PRs · Parallel Linear Mixed Model

[swagger](#) R · author · 9 PRs · Dynamically Generates Documentation from a 'Swagger' Compliant API

[leaflet.providers](#) R · maintainer · 8 PRs · Leaflet Providers

[autocogs](#) R · author · 8 PRs · Automatic Cognition Summaries

[shinychat](#) R, Python · author · 8 PRs · Chat UI Component for 'shiny'

[lazyrmd](#) R · author · 7 PRs · Render R Markdown Outputs Lazily

[leaflet.extras](#) R · author · 7 PRs · Extra Functionality for 'leaflet' Package

[gqlr](#) R · maintainer · 7 PRs · 'GraphQL' Server in R

[packagedocs](#) R · author · 6 PRs · Build Website of Package Documentation

[sass](#) R · author · 6 PRs · Syntactically Awesome Style Sheets ('Sass')

[querychat](#) R, Python · author · 5 PRs · Filter and Query Data Frames in 'shiny' Using an LLM Chat Interface

[leaflet.esri](#) R · author · 4 PRs · 'ESRI' Bindings for the 'leaflet' Package

[gt](#) R · author · 3 PRs · Easily Create Presentation-Ready Display Tables

Also contributed to: [bslib](#), [shinylive](#), [htmltools](#), [posit-sdk](#), [shinytest](#), [httpuv](#), [meetupr](#), [rmarkdown](#), [chatlas](#), [websocket](#), [flexdashboard](#), [webshot2](#), [cachem](#), [later](#), [vetiver](#), [testthat](#), [trelliscope](#), [webshot](#), [webdriver](#), [fastmap](#), [narwhals](#), [plumber2](#), [graphql](#), [ggplot2](#), [redoc](#), [shinyvalidate](#), [jquerylib](#), [routr](#), [brand.yml](#).

## Earlier research software

|                                                                                                                                     |             |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <a href="#">rbokeh</a> , Purdue University · R interface to the Bokeh visualization library.                                        | 2016        |
| <a href="#">DVL</a> , Metamarkets · Functionally reactive JavaScript library binding data to visualizations.                        | 2011 – 2012 |
| <a href="#">helpr</a> , Rice University · Friendlier HTML documentation for R, with search across packages.                         | 2010        |
| <a href="#">cranvas</a> , Iowa State University · Interactive plotting canvas for R, drawing a million points in under a second.    | 2010        |
| <a href="#">tourr</a> , Iowa State University · Geodesic interpolation and basis generation for tour methods in R.                  | 2009        |
| <a href="#">DescribeDisplay</a> , Iowa State University · Publication-quality R graphics from GGobi plots.                          | 2009        |
| <a href="#">geozoo</a> , Iowa State University · Library of high-dimensional geometric objects, including the new Hyper Ring Torus. | 2007        |

## Publications

Gökalp Yavuz, F., and **B. Schloerke** (2020). [Parallel computing in linear mixed models](#). *Computational Statistics* 35(3): 1273–1289.

**Schloerke, B.**, H. Wickham, D. Cook, and H. Hofmann (2016). [Escape from Boxland: Generating a Library of High-Dimensional Geometric Shapes](#). *The R Journal* 8(2): 243–257.

Emerson, J., W. Green, **B. Schloerke**, J. Crowley, D. Cook, H. Hofmann, and H. Wickham (2013). [The Generalized Pairs Plot](#). *Journal of Computational and Graphical Statistics* 22(1).

Hofmann, H., D. Cook, C. Kielion, **B. Schloerke**, J. Hobbs, A. Loy, L. Mosley, D. Rockoff, Y. Huang, D. Wrolstad, and T. Yin (2011). [Delayed, Canceled, on Time, Boarding... Flying in the USA](#). *Journal of Computational and Graphical Statistics* 20(2).

Mosley, L., D. Cook, H. Hofmann, C. Kielion, and **B. Schloerke** (2010). [Monitoring the 2008 Election Visually](#). *Chance* 23(3).

## Talks & workshops

|                                                                                                                                                                |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <a href="#">Beyond Bootstrap: Building Custom Shiny UI with React</a> · posit::conf(2026)                                                                      | Sep 2026 |
| <a href="#">OpenTelemetry in R: Capturing Traces, Logs, and Metrics for Shiny Applications</a> · Open Source in Pharma                                         | Apr 2026 |
| <a href="#">{GGally}: Success and Long-Haul Maintenance Strategies</a> · ggplot2 extenders · <a href="#">video</a>                                             | Nov 2025 |
| <a href="#">Observability at scale: Monitoring Shiny Applications with OpenTelemetry</a> · posit::conf(2025) · <a href="#">video</a>                           | Sep 2025 |
| <a href="#">{plumber2}: Streamlining Web API Development in R</a> · useR! 2025 · <a href="#">video</a>                                                         | Aug 2025 |
| <a href="#">{querychat}: Natural language interfaces for interactive data graphics in Shiny</a> · JSM 2025                                                     | Aug 2025 |
| <a href="#">AI Kung-Fu : Training your LLM client in two function calls</a> · ShinyConf 2025                                                                   | Apr 2025 |
| <a href="#">{shinylive}: Serverless Shiny applications workshop. An exercise in deploying your app to GitHub Pages</a> · R/Pharma 2024 · <a href="#">video</a> | Oct 2024 |
| <a href="#">Data frames in Shiny for Python</a> · <a href="#">video</a>                                                                                        | Aug 2024 |
| <a href="#">{shinylive}: Serverless Shiny apps</a> · ShinyConf 2024 · <a href="#">video</a>                                                                    | Apr 2024 |
| <a href="#">Lessons learned from testing 2500+ Shiny Apps every day</a> · ShinyConf 2023 · <a href="#">video</a>                                               | Mar 2023 |
| <a href="#">{shinytest2}: Regression testing Shiny applications</a>                                                                                            | Oct 2022 |
| <a href="#">{shinytest2}: Regression testing Shiny applications</a> · rstudio::conf(2022) · <a href="#">video</a>                                              | Jul 2022 |
| <a href="#">{shinytest2}: Regression Testing for Shiny Applications</a> · R/Pharma 2022                                                                        | May 2022 |
| <a href="#">shinytest2: Regression Testing for Shiny Applications</a>                                                                                          | Apr 2022 |
| <a href="#">plumber + future: Async Web APIs</a> · Harvard R User Group · <a href="#">video</a>                                                                | Aug 2021 |
| <a href="#">An intro to plumber</a> · Harvard R User Group · <a href="#">video</a>                                                                             | Aug 2021 |
| <a href="#">plumber + future: Async Web APIs</a> · rstudio::global(2021) · <a href="#">video</a>                                                               | Jan 2021 |
| <a href="#">Expanding R Horizons: Integrating R with Plumber APIs</a> · RStudio Webinar · <a href="#">video</a>                                                | Oct 2020 |
| <a href="#">Shiny Developer Series #14: Shining a Light on learnr</a>                                                                                          | Oct 2020 |
| <a href="#">Shiny Developer Series #13: Inside Plumber 1.0</a>                                                                                                 | Sep 2020 |
| <a href="#">Shiny Developer Series #12: reactlog</a>                                                                                                           | Sep 2020 |
| <a href="#">Reactlog 2.0: Debugging the State of Shiny</a>                                                                                                     | Apr 2020 |
| <a href="#">Reproducible Shiny apps with shinymeta</a> · ABACUS 2019                                                                                           | Nov 2019 |
| <a href="#">Debug Large Data Apps: shiny + reactlog + sparklyr</a>                                                                                             | May 2019 |
| <a href="#">Reactlog 2.0: Debugging the State of Shiny</a> · rstudio::conf(2019) · <a href="#">video</a>                                                       | Jan 2019 |
| <a href="#">nasa shiny lightning</a>                                                                                                                           | Aug 2018 |
| <a href="#">Cognostics: Metrics enabling detailed interactive visualization of big data web scraping</a>                                                       | Apr 2017 |
| <a href="#">tidyverse</a>                                                                                                                                      | Mar 2017 |
| <a href="#">Cognostics: Metrics enabling detailed interactive visualization of big data</a>                                                                    | Jan 2017 |
| <a href="#">User2016!</a> · useR! 2016                                                                                                                         | Aug 2016 |
| <a href="#">trelliscope</a>                                                                                                                                    | Jun 2016 |
|                                                                                                                                                                | Mar 2016 |

|                                                                                                                              |          |
|------------------------------------------------------------------------------------------------------------------------------|----------|
| gqlr = GraphQL + R: a data query language and runtime · WOMBAT                                                               | Feb 2016 |
| web scraping                                                                                                                 | Oct 2015 |
| Large Data Visualization for Fields Institute's THEMATIC PROGRAM ON STATISTICAL INFERENCE, LEARNING, AND MODELS FOR BIG DATA | Feb 2015 |
| ggplot2 spatial statistics                                                                                                   | Oct 2014 |
| Git                                                                                                                          | Jun 2014 |
| minimize work inefficiencies                                                                                                 | Feb 2014 |

## Videos

|                                                                                 |          |
|---------------------------------------------------------------------------------|----------|
| Git Worktree-First Workflow + Conductor                                         | Aug 2026 |
| Shiny 1.2 Release: Wrap your data frame with Narwhals!                          | Oct 2024 |
| Getting Started with {shinytest2} Part I · Example + basics                     | Jul 2022 |
| Getting Started with {shinytest2} Part 3 · Using shiny.testmode in {shinytest2} | Jul 2022 |
| Getting Started with {shinytest2} Part 2 · Exporting values                     | Jul 2022 |
| Programming Games with Shiny · Roll the Dice: with Quosures!                    | Jun 2022 |
| Programming Games with Shiny · Roll the Dice                                    | May 2022 |
| Programming Games with Shiny · Guess the Number                                 | May 2022 |
| Programming Games with Shiny · Dragon Realm                                     | Mar 2022 |
| Maximize computing resources using future_promise()                             | Feb 2022 |
| {reactlog} Rundown                                                              | Feb 2022 |

## Blog posts

|                                                           |          |
|-----------------------------------------------------------|----------|
| Introducing shinyreact: React UI backed by a Shiny server | Sep 2026 |
| Shiny for Python 1.8                                      | Sep 2026 |
| Shiny updates: R 1.14, Python 1.7, bslib 0.12             | Aug 2026 |
| Bringing OpenTelemetry to R in production                 | May 2026 |
| Shiny for Python 1.6 brings toolbars and OpenTelemetry    | Apr 2026 |
| OpenTelemetry + Shiny for R v1.12                         | Dec 2025 |
| Tool Calling UI in shinychat                              | Nov 2025 |
| promises v1.5.0                                           | Nov 2025 |
| Shiny at posit::conf(2025)                                | Sep 2025 |
| Posit at SciPy 2025                                       | Jul 2025 |
| Shiny 1.4 brings bookmarking and Generative AI docs       | Apr 2025 |
| Shiny for Python 1.2.0                                    | Oct 2024 |
| Testing Shiny Apps in Python                              | Oct 2024 |
| Announcing Shiny for Python 1.0                           | Jul 2024 |
| Editable data tables in Shiny for Python 0.9.0            | May 2024 |
| Introducing: Shiny Express                                | Jan 2024 |
| Shiny for Python 0.6.1                                    | Dec 2023 |
| Shiny for Python 0.6.0                                    | Nov 2023 |
| Shiny for Python 0.5.0                                    | Aug 2023 |
| Shiny for Python 0.4.0                                    | Jun 2023 |
| Shiny for Python out of alpha                             | Apr 2023 |
| plumber 1.1.0                                             | Mar 2021 |
| Shiny 1.6                                                 | Feb 2021 |
| learnr 0.10.0                                             | Dec 2019 |
| leaflet 2.0.0                                             | May 2018 |

## Honors & awards

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|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <a href="#">National Science Foundation Graduate Research Fellowship</a> · the only Computational Statistics awardee of 13 statistics recipients in 2012 | 2012 – 2017 |
| Visiting Scientific Researcher, Monash University · large data exploration tools with NUMBAT                                                             | 2016        |
| <a href="#">Google Summer of Code</a> , <a href="#">GGally::ggduo</a> · pairs plots for two groups of variables                                          | 2016        |
| Best Dynamic / Interactive Visualization, Data Insight SF                                                                                                | 2011        |
| NSF Graduate Research Fellowship, honorable mention                                                                                                      | 2011        |
| <a href="#">ASA Data Expo</a> , 2nd place (Iowa State working group)                                                                                     | 2009        |

## Teaching & mentoring

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Internship Mentor</b> , Posit · Samuel Bharti: a <a href="#">bioinformatics Shiny showcase</a> , 10 Shiny apps built on 5 packages                                                                                                                                                | 2026        |
| <b>Workshop Teaching Assistant</b> , <a href="#">rstudio::conf</a> / <a href="#">posit::conf</a> · <a href="#">Shiny in Production</a> (2019), <a href="#">Applied Machine Learning</a> (2020), <a href="#">Level Up with Shiny for R</a> (2024), <a href="#">Shiny for R</a> (2025) | 2019 – 2025 |
| <b>Internship Mentor</b> , RStudio · Leslie Huang: <a href="#">leaflet.providers</a> and education materials                                                                                                                                                                         | 2019        |
| <b>Internship Advisor</b> , RStudio · Daniel Chen: <a href="#">gradethis</a> , a companion package for grading <a href="#">learnr</a> exercises                                                                                                                                      | 2019        |
| <b>Instructor</b> , <a href="#">R for Data Science Seminar</a> , <a href="#">Purdue University</a>                                                                                                                                                                                   | 2017        |
| <b>Teaching Assistant</b> , Big Data Analytics: Statistics and Data Visualization, Purdue University                                                                                                                                                                                 | 2017        |
| <b>Teaching Assistant</b> , Hadley Wickham's R Master Class, San Francisco and Chicago                                                                                                                                                                                               | 2011 – 2015 |

## Service

|                                                                                                                                                                                    |             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>President</b> , Statistics Graduate Student Office, Purdue University · Ran a town hall, a graduate student mentor program, and an esteemed speaker event with Arthur Dempster. | 2016 – 2017 |
| <b>Graduate Student Mentor</b> , Purdue University Statistics                                                                                                                      | 2014 – 2016 |
| <b>StatCom Assistant</b> , Purdue University                                                                                                                                       | 2014 – 2016 |

## Skills

|              |                                                                     |
|--------------|---------------------------------------------------------------------|
| Languages    | R, Python, TypeScript, JavaScript, HTML, CSS, SQL, Bash             |
| Frameworks   | Shiny (R and Python), React, plumber, htmltools, ggplot2, tidyverse |
| Testing & CI | Playwright, pytest, testthat, shinytest2, GitHub Actions            |
| Publishing   | CRAN, PyPI, npm, conda-forge, pkgdown, Quarto, GitHub Pages         |